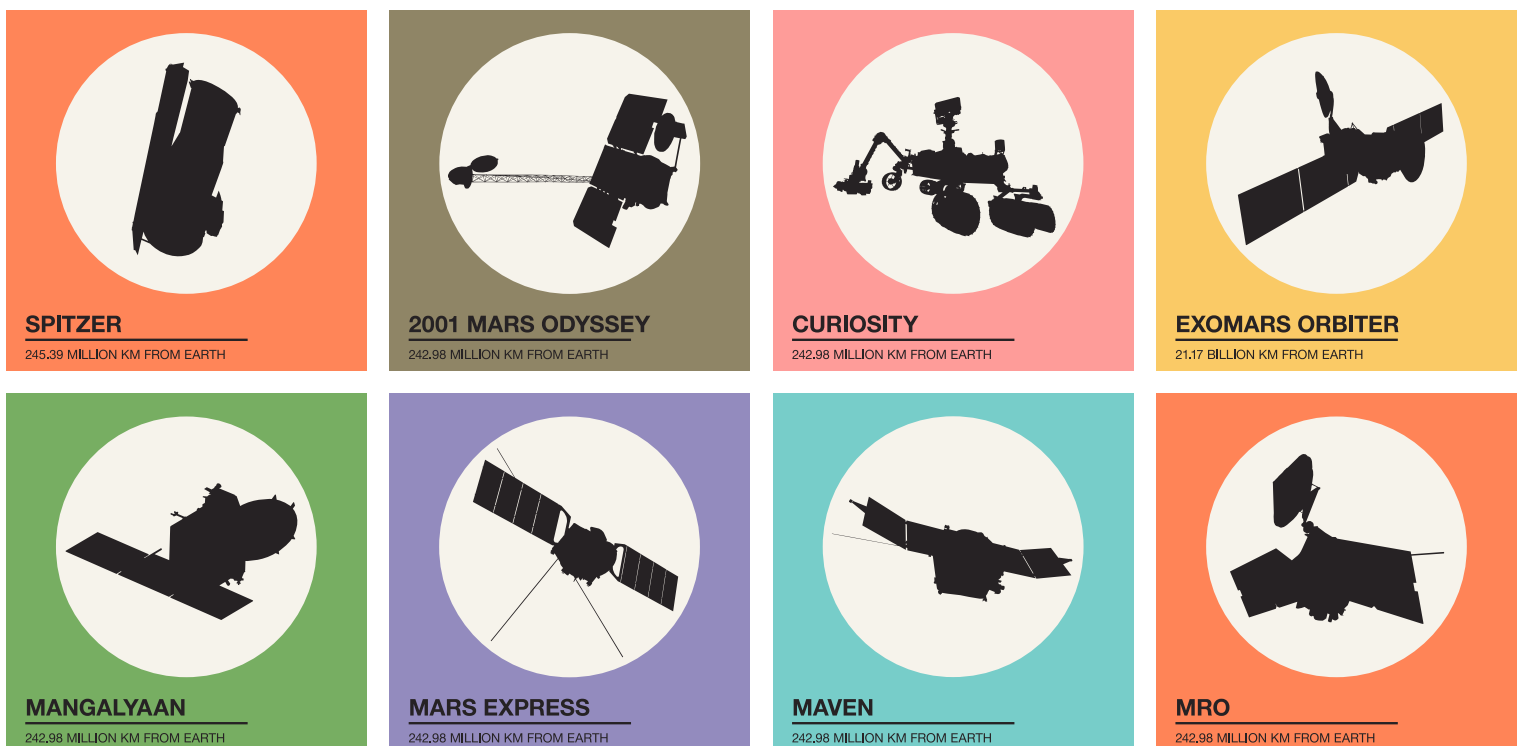




Beyond Earth

A CHRONICLE OF DEEP SPACE EXPLORATION, 1958–2016

Asif A. Siddiqi



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The artwork titled Spaceprob.es is a companion piece to the Web site that catalogs the active human-made machines that freckle our solar system. Each space probe's silhouette has been paired with its distance from Earth via the Deep Space Network or its last known coordinates.



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Firsts

in the History of Deep Space Exploration

Absolute Firsts

First attempt to launch a probe into deep space:
USA / **Able 1** [**Pioneer 0**] / 17 August 1958

First probe to reach escape velocity:
USSR / **Soviet Space Rocket** [**Luna 1**] /
2 January 1959

First spacecraft to impact on another celestial body:
USSR / **Second Soviet Space Rocket**
[**Luna 2**] / 14 September 1959 (Moon)

First successful planetary mission:
USA / **Mariner II** / 14 December 1962 (Venus)

First spacecraft to impact another planet:
USSR / **Venera 3** / 1 March 1966 (Venus)

First spacecraft to make a survivable landing on a
celestial body:
USSR / **Luna 9** / 3 February 1966 (Moon)

First spacecraft to orbit a celestial body other than
Earth or the Sun:
USSR / **Luna 10** / 2 April 1966 (Moon)

First successful planetary atmospheric entry probe:
USSR / **Venera 4** / 18 October 1967 (Venus)

First liftoff from another celestial body:
USA / **Surveyor VI** / 17 November 1967 (Moon)

First transmission from the surface of another planet:
USSR / **Venera 7** / 15 December 1970 (Venus)

First robotic spacecraft to recover and return samples
from another celestial body:
USSR / **Luna 16** / 12–21 September 1970 (Moon)

First wheeled vehicle on another celestial body:
USSR / **Lunokhod 1** / 17 November 1970 (Moon)

First spacecraft to fly through the asteroid belts:
USA / **Pioneer 10** / out in February 1973

First spacecraft to use gravity assist to change its
interplanetary trajectory:
USA / **Mariner 10** / 5 February 1974 (at Venus)

First spacecraft to fly past multiple planets:
USA / **Mariner 10** / 5 February 1974 (Venus)
and 29 March 1974 (Mercury)

First spacecraft to transmit photos from the surface of
another planet:
USSR / **Venera 10** / 22 October 1975 (Venus)